

Computer-Aided Dispatch (CAD) System

Software Requirements Specification

Ambulance / EMS Dispatch Platform

Version 1.0

August 2026

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for a Computer-Aided Dispatch (CAD) system supporting ambulance and Emergency Medical Services (EMS) operations. It defines the system's user roles, required capabilities per role, underlying architecture, and key operational workflows, and serves as the baseline reference for design and implementation.

1.2 Scope

The system covers the full incident lifecycle: emergency call intake, unit dispatch and routing, real-time field status updates, hospital hand-off, fleet health monitoring, and administrative oversight. It is designed as a system-design project built on an open-source technology stack, and is intended to demonstrate a realistic, production-style CAD architecture rather than to replace a certified public-safety system.

1.3 Intended Audience

- Project evaluators / instructors reviewing the system design
- Developers implementing frontend, backend, and integration components
- Anyone assessing the system's operational and role-based requirements

1.4 Definitions & Acronyms

Term	Definition
CAD	Computer-Aided Dispatch — software used to manage emergency incidents and coordinate response units.
EMS	Emergency Medical Services.
ETA	Estimated Time of Arrival.
OSRM	Open Source Routing Machine — road-network routing engine used for distance/ETA calculation.
MQTT	Message Queuing Telemetry Transport — lightweight publish/subscribe messaging protocol.
CAN bus	Controller Area Network — in-vehicle network used for diagnostic/telemetry data.
PostGIS	Spatial extension for PostgreSQL enabling geospatial queries.
Unit	An ambulance vehicle and its assigned crew, treated as a single dispatchable resource.

2. System Overview

The CAD system connects seven categories of users around a shared, real-time view of incidents and ambulance resources. A Call Taker logs an emergency and hands it to a Dispatcher, who assigns the nearest suitable Unit using road-network-aware routing. The assigned Paramedic crew receives the dispatch on an in-vehicle/mobile client, updates status through the response lifecycle, and hands the patient off to a Hospital,

which is notified in advance of the incoming patient. An Operations Supervisor monitors performance and reallocates resources fleet-wide, an Admin configures and governs the system, and Fleet/Vehicle Maintenance Staff keep the ambulance fleet roadworthy using live vehicle diagnostics.

3. User Roles and Personas

The system defines seven user roles, summarized below. A single person may hold more than one role (e.g. a supervisor who is also a dispatcher) if the deployment requires it.

Role	Description
Call Taker	Front-line staff who answer emergency calls, capture caller and incident information, and triage priority before handoff to dispatch.
Dispatcher	Assigns and coordinates ambulance units to incidents, monitors live unit status/location, and manages the active incident queue.
Admin	Configures system-wide settings: user accounts, roles/permissions, stations, vehicle master data, and reviews audit logs.
Paramedic	Field crew member operating the in-vehicle/mobile client; receives dispatch, updates unit status, and closes out incidents.
Operations Supervisor	Oversees fleet-wide operations, monitors KPIs, reallocates resources during surges, and reviews post-incident performance.
Hospital	Receiving facility staff who get pre-arrival notifications, view incoming ambulance queue, and report bed/capacity availability.
Fleet / Vehicle Maintenance Staff	Maintains vehicle roadworthiness; monitors CAN bus diagnostics, schedules service, and sets vehicles in/out of service.

4. Functional Requirements

Requirements are grouped by the primary role that uses them and prioritized using MoSCoW (Must / Should / Could).

4.1 Call Taker

ID	Requirement	Priority
FR-CT-01	Capture a new emergency call with caller name/callback number, free-text narrative, and chief complaint.	Must
FR-CT-02	Determine incident location via address lookup, map pin-drop, or automatic number identification (ANI)/caller-ID lookup where available.	Must
FR-CT-03	Apply a priority/triage level to the incident using a configurable protocol (e.g. Echo/Delta/Charlie/Bravo/Alpha).	Must
FR-CT-04	Submit the incident to the dispatch queue in real time upon call completion.	Must
FR-CT-05	Flag duplicate or related calls against an existing open incident within a configurable radius/time window.	Should

ID	Requirement	Priority
FR-CT-06	Attach supplementary notes to an incident already in progress (e.g. caller calls back).	Should

4.2 Dispatcher

ID	Requirement	Priority
FR-DP-01	View a live, map-based display of all units with current status (available, en route, on scene, transporting, at hospital, out of service).	Must
FR-DP-02	View an incident queue ordered by priority and time received.	Must
FR-DP-03	Receive a system-recommended nearest-available-unit suggestion computed via road-network routing (not straight-line distance).	Must
FR-DP-04	Manually override the recommended unit and assign any available unit to an incident.	Must
FR-DP-05	Transmit dispatch details (location, priority, patient info) to the assigned unit in real time over the messaging bus.	Must
FR-DP-06	Reassign or escalate an incident (e.g. request a second unit, upgrade priority) after initial dispatch.	Must
FR-DP-07	View estimated time of arrival (ETA) for the assigned unit, recalculated as the unit moves.	Should
FR-DP-08	Send free-text or canned messages to a unit crew during an active incident.	Should

4.3 Admin

ID	Requirement	Priority
FR-AD-01	Create, edit, deactivate user accounts and assign one or more of the defined roles.	Must
FR-AD-02	Define and edit role-based permission sets (module-level access control).	Must
FR-AD-03	Maintain master data for stations, response zones, and vehicle/unit records.	Must
FR-AD-04	View a searchable, immutable audit log of key actions (dispatch changes, status overrides, account changes).	Must
FR-AD-05	Configure triage protocols, priority levels, and dispatch-recommendation parameters.	Should
FR-AD-06	Generate operational reports (response times, incident volumes, unit utilization) and export to CSV/PDF.	Should

4.4 Paramedic

ID	Requirement	Priority
FR-PM-01	Receive a dispatch notification on the in-vehicle/mobile client with incident location, priority, and available patient details.	Must
FR-PM-02	Update unit status through the response lifecycle (en route → on scene → transporting → at hospital → available).	Must

ID	Requirement	Priority
FR-PM-03	View turn-by-turn navigation guidance to the incident location and, subsequently, to the destination hospital.	Must
FR-PM-04	Select a destination hospital, optionally guided by hospital-reported capacity/diversion status.	Should
FR-PM-05	Submit an incident closeout with outcome summary and patient handoff confirmation.	Must
FR-PM-06	Continue to operate (queue status updates) during brief connectivity loss and sync once reconnected.	Should

4.5 Operations Supervisor

ID	Requirement	Priority
FR-OS-01	View a fleet-wide operational dashboard: unit availability counts, open incidents, and average response time.	Must
FR-OS-02	Reallocate units across zones/stations manually during demand surges or unit shortages.	Must
FR-OS-03	Review post-incident performance metrics against configurable response-time targets.	Should
FR-OS-04	Manage crew/vehicle shift rosters and view current on-duty staffing by station.	Should
FR-OS-05	Receive alerts when zone coverage falls below a configurable minimum-unit threshold.	Could

4.6 Hospital

ID	Requirement	Priority
FR-HOS-01	Receive a pre-arrival notification for an inbound unit, including patient condition summary and ETA.	Must
FR-HOS-02	View a queue of all ambulances currently inbound to the facility.	Must
FR-HOS-03	Report and update current bed/capacity status and diversion state, visible to dispatch and paramedic clients.	Must
FR-HOS-04	Acknowledge receipt of an inbound patient notification.	Should

4.7 Fleet / Vehicle Maintenance Staff

ID	Requirement	Priority
FR-FM-01	View real-time vehicle diagnostic/fault data streamed from the on-board CAN bus interface.	Must
FR-FM-02	Receive automated alerts when a fault code or telemetry threshold (e.g. engine temperature, mileage) is exceeded.	Must
FR-FM-03	Schedule and log preventive/corrective maintenance against a specific vehicle.	Must

ID	Requirement	Priority
FR-FM-04	Set a vehicle's operational status to out-of-service or return it to in-service, which propagates to dispatch availability.	Must
FR-FM-05	View maintenance history and mileage/engine-hours per vehicle.	Should

5. System Architecture & Technology Stack

The system is built on an open-source stack chosen to demonstrate realistic CAD architecture end to end.

Component	Technology	Role in the system
Frontend (web)	React + Leaflet	Dispatcher, admin, operations supervisor, and hospital web consoles; live map rendering of units and incidents.
Backend / API	Python (FastAPI)	Business logic, REST endpoints, authentication/authorization, orchestration of routing requests.
Spatial database	PostgreSQL + PostGIS	Persistent storage of incidents, units, stations, hospitals, and geospatial queries (nearest-unit, zone coverage).
Routing engine	OSRM over OpenStreetMap data	Road-network distance/ETA calculation for unit recommendation and navigation guidance.
Real-time messaging	Mosquitto (MQTT)	Low-latency publish/subscribe channel for unit status, dispatch pushes, and location telemetry.
Vehicle telemetry	Ambulance Simulator	Simple ambulance simulator that mimics the real ambulance behavior following routes to the incident location

5.1 Notes on real-time design

- Unit location and status changes are published over MQTT so all subscribed clients (dispatcher map, hospital queue, ops dashboard) update without polling.
- Nearest-unit recommendations and navigation both call the same OSRM routing service so dispatcher ETAs and in-vehicle guidance stay consistent.
- CAN bus telemetry is ingested as a separate stream feeding the fleet/maintenance module, decoupled from the dispatch-critical path.

5.2 Data Flow Diagram

The diagram below traces how data moves between actors, core services, and data stores across the dispatch, field-response, hospital hand-off, and fleet-telemetry paths described above.

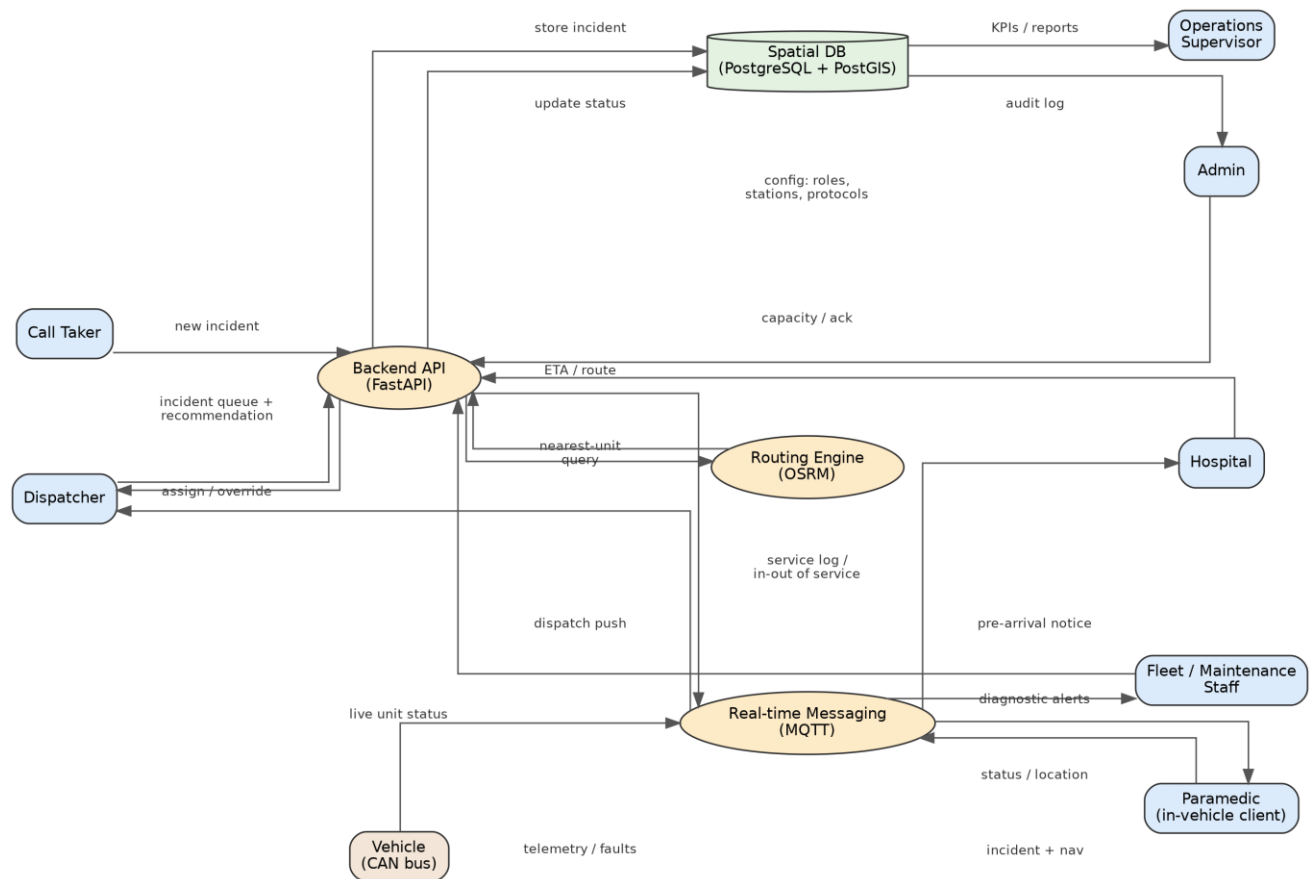


Figure 1. CAD system data flow — actors and processes exchange data through the Backend API, OSRM routing engine, MQTT messaging bus, and the PostGIS-backed spatial database.

6. Data Model (Key Entities)

Entity	Description	Key Attributes
Incident	A reported emergency event.	incident_id, location (geometry), priority, status, created_at, chief_complaint, caller_ref
Call	A logged phone contact tied to an incident.	call_id, incident_id, caller_number, narrative, taken_by (call taker)
Unit / Vehicle	A physical ambulance and its current operating state.	unit_id, call_sign, current_location, status, assigned_crew, vehicle_health_status
Dispatch	The assignment of a unit to an incident.	dispatch_id, incident_id, unit_id, assigned_at, eta, dispatcher_id
Personnel / User	Any system user, with one or more roles.	user_id, name, role(s), station_id, credentials
Station	A physical base from which units operate.	station_id, name, location, coverage_zone

Entity	Description	Key Attributes
Hospital	A receiving facility.	hospital_id, name, location, capacity_status, diversion_flag
Maintenance Record	A service event tied to a vehicle.	record_id, unit_id, type, date, notes, odometer, fault_codes
Roster / Shift	Scheduled staffing for a station/unit.	roster_id, unit_id, crew_ids, shift_start, shift_end

7. Key Workflow: End-to-End Dispatch

- **1.** Call Taker receives an emergency call, logs caller and location details, and assigns a triage priority.
- **2.** The incident is submitted and immediately appears in the Dispatcher's prioritized incident queue.
- **3.** The system queries the routing engine (OSRM) for the nearest available unit(s) by road-network travel time, not straight-line distance.
- **4.** Dispatcher confirms or overrides the recommended unit and assigns it to the incident.
- **5.** The dispatch is pushed to the Paramedic's in-vehicle client over MQTT in real time, along with navigation guidance.
- **6.** Paramedic updates status as the response progresses: en route → on scene → transporting → at hospital.
- **7.** As the unit transports a patient, the destination Hospital receives a pre-arrival notification with patient summary and ETA.
- **8.** Hospital acknowledges the notification and may report capacity/diversion status back into the system.
- **9.** Paramedic closes out the incident with an outcome summary upon patient handoff; the unit returns to Available status.
- **10.** Operations Supervisor and Admin can review the full incident timeline, response-time metrics, and any status changes via the audit log and reporting module.
- **11.** Fleet/Maintenance Staff continuously receive CAN bus telemetry in the background; a fault code can trigger an automatic out-of-service flag that removes the unit from the dispatch pool until cleared.

8. Non-Functional Requirements

Category	Requirement
Performance	Dispatch recommendation and unit-status updates must be delivered to relevant clients within 2 seconds under normal load.
Availability	Core dispatch path (call intake → assignment → unit notification) must target 99.9% uptime; system should degrade gracefully, not fail closed.
Scalability	Architecture must support horizontal scaling of API and messaging layers as unit/station count grows.
Security	All access is authenticated and role-based; sensitive patient data is encrypted in transit (TLS) and at rest.

Category	Requirement
Data Privacy	Patient-identifiable information is restricted to roles that require it (call taker, dispatcher, paramedic, receiving hospital).
Reliability of Location Data	Geospatial routing and ETA calculations must use actual road-network data, not straight-line distance.
Auditability	Every dispatch decision, status change, and administrative action is logged with actor, timestamp, and prior/new value.
Resilience / Offline Tolerance	Field (paramedic) clients must tolerate brief network loss without losing in-progress status updates.
Interoperability	System is built on open standards (REST/MQTT, GeoJSON, standard mapping tiles) to ease integration with hospital or regional systems.
Usability	Dispatcher and call-taker interfaces must support high-stress, time-critical use: minimal clicks to dispatch, large legible status indicators.

9. Role / Module Access Matrix

“Full” denotes create/edit/delete access; “View” denotes read-only access; “—” denotes no access. Columns, left to right: Call Taker, Dispatcher, Admin, Paramedic, Operations Supervisor, Hospital, Fleet/Maintenance.

Module	CT	DP	AD	PM	OS	HOS	FM
Incident Intake	Full	View	—	—	View	—	—
Dispatch & Unit Assignment	—	Full	View	Status only	Full	—	—
Live Map / Fleet View	—	Full	View	—	Full	—	View (fleet health)
User & Role Management	—	—	Full	—	—	—	—
Station / Vehicle Master Data	—	View	Full	—	View	—	Edit (vehicle data)
Audit Log	—	—	Full	—	View	—	—
Reports & KPIs	—	View	Full	—	Full	—	View (maintenance)
Bed / Capacity Status	—	View	—	View	View	Full	—
Maintenance & CAN Diagnostics	—	—	View	—	View	—	Full
Roster / Shift Management	—	—	View	View (own)	Full	—	—

10. Assumptions & Constraints

- The system is deployed within a single EMS region/agency; multi-agency federation is out of scope for this version.
- Map and routing data rely on OpenStreetMap coverage and accuracy for the deployment area.
- Units are assumed to have a persistent connection for MQTT telemetry, with brief offline periods tolerated but not extended outages.
- Hospital capacity data is self-reported by hospital staff; the system does not integrate with hospital EHR systems in this version.
- CAN bus data access assumes a compatible on-board diagnostic interface is present on each vehicle.

11. Out of Scope / Future Enhancements

- Automated (fully algorithmic) unit dispatch without dispatcher confirmation
- Direct integration with hospital electronic health record (EHR) systems
- Multi-agency / cross-jurisdiction resource sharing
- Native mobile applications (initial paramedic/hospital clients are web-based)
- Predictive demand modeling for proactive unit repositioning

12. Document Control

Version	Date	Description
1.0	August 2026	Initial specification covering all seven user roles, functional/non-functional requirements, architecture, and workflows.